wide variety of different kinds of Gridworld environments of varying complexity and difficulty. Some of these Gridworld environments are so difficult (largely due to sparse rewards) that only the most cutting-edge reinforcement learning algorithms are capable of making headway. Install the package using `pip`:

```
>>> pip3 install gym-minigrid
```

We will be using a somewhat difficult environment in which the Gridworld agent must navigate to a key, pick it up, use it to open a door, and then navigate to a goalpost in order to receive a positive reward (figure 10.19). This is a lot of steps before it ever sees a reward, so we will encounter the sparse reward problem. This would actually be a great opportunity to employ curiosity-based learning, but we will restrict ourselves to the smallest version of the grid, the MiniGrid, so that even a random agent would eventually find the goal, so we can successfully train without curiosity. For the larger grid variants of this environment, curiosity or related approaches would be almost necessary.

Figure 10.19. The MiniGrid-DoorKey environment. In this environment, the agent (the triangle) must first navigate to the key, pick it up, navigate to the door (the hollow square), open it, and then navigate to the solid square. Each game initializes the objects on the grid randomly, and the agent only has a partial view of the grid indicated by the highlighted region around it.